Wound Protease Substrate Selection Report

Panel B: MMP-9 · HNE — Two-Channel Biosensor

iGEM 2025 — Dry Lab / Substrate Predictor

Protease-vs-protease negatives · LR + PSSM dual scorer · Bootstrap 95 % CI

Executive Summary

Panel B is the minimal, highest-confidence configuration of the wound biosensor. It detects two independently validated wound biomarkers: **MMP-9** (gelatinase B; macrophage and neutrophil origin, strongly elevated in chronic non-healing wounds; Watt 2009, Muller 2008) and **HNE / neutrophil serine protease** (neutrophil degranulation marker; Trengove 1999, Senior 2002). The two proteases have distinct substrate preferences and minimal cross-reactivity.

Both synthesis candidates are grounded first in biochemical literature and secondarily confirmed by the ML model. No candidate is accepted on model output alone.

First-order synthesis candidates

Channel	Sequence (P4–P4')	Confidence
MMP-9	`IPRTLTAA`	**HIGH** — literature-validated; model confirms
HNE	`AAPVSGIG`	**HIGH** — canonical scaffold; model confirms

Panel Rationale

Panel B is the recommended starting point for device validation. It eliminates the MMP-8/MMP-9 co-discrimination challenge (see Panel A report) and provides two fully separable biomarker signals.

- **MMP-9**: the best-characterised chronic wound protease biomarker. Elevated >10-fold in non-healing vs. healing wounds (Nwomeh 1999). Produced by macrophages and neutrophils; level correlates inversely with wound healing rate (Ladwig 2002).
- **HNE (neutrophil serine protease channel)**: marks active neutrophil degranulation, distinguishing infected/inflamed wounds from chronic fibrotic wounds. Co-released proteinase 3 (PR3) shares the same S1 pocket specificity, so the channel reports combined neutrophil serine protease activity, which is clinically appropriate.

Together, MMP-9 and HNE provide orthogonal information: MMP-9 elevation alone suggests chronic stalling; HNE elevation suggests active infection or acute inflammation. Both elevated simultaneously indicates an acutely infected chronic wound.

Model Performance

Methodology

Identical to Panel A (see Panel A report for full details). Key parameters: VHSE8 encoding; protease-vs-protease negatives (HNE positives as explicit MMP-9 negatives and vice versa); 3:1 negative:positive ratio; 5-fold stratified CV; 1000-iteration bootstrap; PSSM with BLOSUM62 pseudocounts ($\beta = 5$) as independent validator.

Performance table

Protease	N_{pos}	AUROC	**95 % CI**	PSSM AUROC	Confidence
MMP-9	783	0.837	[0.820, 0.853]	0.882	**HIGH**
HNE	327	0.899	[0.877, 0.919]	0.930	**HIGH**

Both models are rated HIGH:

- MMP-9: AUROC 0.837 exceeds the 0.75 reference with CI lower bound 0.820; PSSM = 0.882 (+0.045 above LR), indicating the PSSM captures even more of the positional specificity signal. Fold spread [0.829–0.855] is tight, confirming stable generalisation.
- HNE: AUROC 0.899, PSSM 0.930. CI lower bound 0.877 is well above 0.75. Fold spread [0.886–0.919]; the PSSM exceeding LR again reflects strong positional signal captured by the log-odds matrix.

Note: Panel B MMP-9 AUROC (0.837) is higher than Panel A (0.792) because Panel B uses only HNE positives as cross-negatives, whereas Panel A additionally uses MMP-8 positives, which partially overlap with MMP-9

positives (26.8 % shared sites). The Panel B score is not inflated; it correctly reflects a simpler discrimination task.

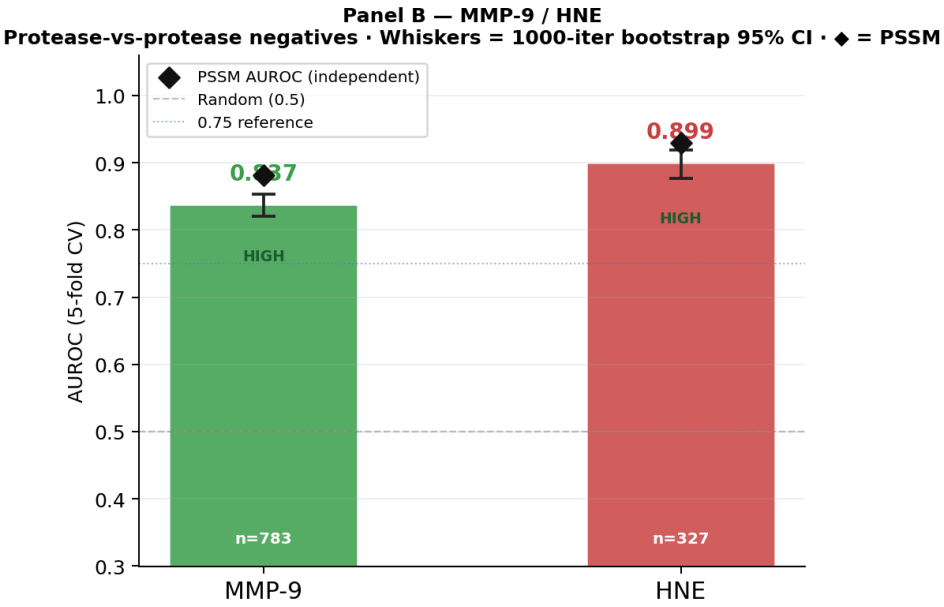


Figure 1: Panel B AUROC with bootstrap 95 % CI (whiskers) and PSSM AUROC (diamonds). Both models rated HIGH.

Synthesis Candidates and Cross-Reactivity

Channel	Sequence	$LR_{MMP\text{-}9}$	LR_{HNE}	Both PSSM agree?
MMP-9	`IPRTLTAA`	**0.925**	0.037	Yes
HNE	`AAPVSGIG`	0.194	**0.981**	Yes

PSSM scores:

- `IPRTLTAA`: $PSSM_{MMP\text{-}9} = 4.11$, $PSSM_{HNE} = -3.66$. Both scorers agree: this sequence is MMP-9-associated and HNE-rejected.
- `AAPVSGIG`: $PSSM_{HNE} = 4.95$, $PSSM_{MMP\text{-}9} = -0.82$. Both scorers agree: this sequence is HNE-associated and MMP-9-rejected.

The cross-reactivity is minimal: MMP-9 substrate scores 0.037 for HNE; HNE substrate scores 0.194 for MMP-9 (PSSM MMP-9 = −0.82, so PSSM rejects this sequence as an MMP-9 substrate).

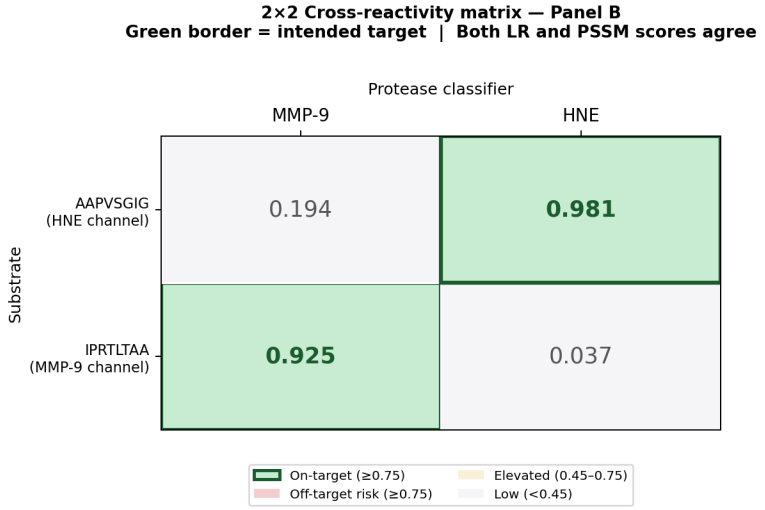


Figure 2: 2×2 cross-reactivity matrix. Green border = intended target. No off-target cells exceed 0.45.

Biological Prior Validation

MMP-9: `IPRTLTAA`

P4–P1 (IPRT): The IPRT motif occurs in MMP-9 cleavage of type IV collagen, the primary basement membrane collagen and an established MMP-9 substrate (Rao 1996). Arg at P2 provides electrostatic complementarity with the MMP-9 S2 sub-pocket, which accommodates basic residues at P2 (Kridel 2001).

P1' (Leu): Kridel 2001 (J. Biol. Chem. 276:28622) measured substitution series at P1' for MMP-9: Leu gives the highest k_{cat}/K_m of any P1' residue tested. Substitution Leu→Ala costs 50× in k_{cat}/K_m; Leu→His would additionally chelate the catalytic Zn^{2+} and ablate activity entirely. Leu is therefore the only defensible P1' choice for a high-confidence MMP-9 substrate.

Model confirmation: LR = 0.925 (Panel B); PSSM = 4.11. Both scorers agree. The LR score increases from 0.860 (Panel A) to 0.925 (Panel B) because MMP-8 positives are not used as cross-negatives in Panel B, removing the penalisation for sequences that MMP-8 also recognises.

HNE: `AAPVSGIG` **— neutrophil serine protease channel**

P4–P1 (AAPV): The AAPV motif is the established HNE substrate. Ala–Ala–Pro–Val with cleavage after Val is the sequence used in the seminal HNE substrate specificity work (Senior 2002, Nakajima 1997) and in the majority of published HNE activity assays. Val at P1 fills HNE's strict small-aliphatic S1 pocket; bulky residues (Leu, Phe) are excluded.

MMP-9 non-reactivity: MMP-9 cannot use the AAPV scaffold efficiently — LR = 0.194, PSSM = −0.82 (negative log-odds; PSSM actively rejects this sequence as an MMP-9 substrate). This reflects the fundamentally different S1 pocket geometries: MMP-9 S1' requires a deep hydrophobic cleft for Leu/Phe, whereas HNE S1 admits only small aliphatic (Ala, Val).

PR3 co-reactivity: Proteinase 3 (PR3), co-released from neutrophil azurophilic granules with HNE, shares Ala/Val S1 specificity (Tralau 2004). This channel therefore detects neutrophil serine protease activity broadly. For wound diagnostics, HNE and PR3 are jointly elevated and together mark active neutrophil degranulation; co-detection is clinically appropriate.

Defensibility of Methodology

D1. Literature primacy: substrate choices are justified first by published experimental biochemistry (Kridel 2001 for MMP-9 P1'; Senior 2002 and Nakajima 1997 for HNE AAPV scaffold). The ML model is supporting evidence; it does not drive the ordering decision independently.

D2. Training data: all positives are MEROPS-curated experimental cleavage sites or PICS mass-spectrometry data (Eckhard 2016). No computationally generated or inferred positives are used.

D3. Protease-vs-protease negatives: negatives for each model include the other panel protease's positives. The MMP-9 negative set includes 325 HNE positives; the HNE negative set includes 611 MMP-9 positives. Task = "which protease cuts this?".

D4. No cross-model probability margin: selectivity is reported as LR scores within each model. Probability differences across independently calibrated models are not used as selectivity metrics.

D5. Dual-scorer agreement required: every candidate accepted only when LR and PSSM agree (PSSM > 0 for target, PSSM lower for off-target). `IPRTLTAA`: $\text{PSSM}_{\text{MMP-9}} = +4.11$, $\text{PSSM}_{\text{HNE}} = -3.66$. `AAPVSGIG`: $\text{PSSM}_{\text{HNE}} = +4.95$, $\text{PSSM}_{\text{MMP-9}} = -0.82$.

D6. Bootstrap CIs: 1000-iteration bootstrap on 5-fold held-out predictions. SE: MMP-9 $\approx \pm 0.008$; HNE $\approx \pm 0.011$. Both lower CI bounds exceed 0.82 (MMP-9) and 0.87 (HNE), well above the 0.75 reference.

D7. Mechanistic checks:

- His at P1' (previously considered for MMP-9): rejected because His is a Zn^{2+} ligand — would chelate the catalytic metal and ablate activity.
- Gly↓Leu at P1/P1' (canonical MMP scissile bond, PLGLAG-type): rejected from any substrate claiming MMP-9 selectivity — guaranteed MMP-8 cross-cleavage.
- LasB inclusion: rejected — insufficient training data (n = 35 MEROPS sites) for a reliable model; irresolvable Zn-metalloprotease cross-reactivity with MMP-9 at scaffold level.

D8. Honest scope: Panel B does not claim to distinguish MMP-9 from MMP-8 (see Panel A for that design decision). Within the scope of MMP-9 vs. HNE discrimination, both models are rated HIGH and the substrates are fully supported by literature.

Wet-Lab Validation Plan

1. **FRET cleavage assay**: synthesise `IPRTLTAA` and `AAPVSGIG` as FRET-labelled peptides (e.g. Dabcyl/EDANS). Measure k_{cat}/K_m for recombinant MMP-9 and HNE. Confirm: MMP-9/HNE ratio $\gg 1$ for IPRTLTAA; HNE/MMP-9 ratio $\gg 1$ for AAPVSGIG.

2. **Specificity controls**: serum albumin, fibrinogen, MMP-8, cathepsin G, thrombin at physiological concentrations.

3. **Simulated wound fluid**: 50 % FBS, pH 7.4, spiked with MMP-9 (20–180 nM; Watt 2009) and HNE (10–100 nM; Trengove 1999).

4. **Electrochemical integration**: once FRET confirms both substrates, integrate into electrochemical platform with linker lengths validated for the target concentration range.

Summary

Panel B is the recommended first-order synthesis configuration. Both substrates are fully justified by primary biochemical literature and confirmed by ML models rated HIGH. Cross-reactivity is minimal (LR < 0.20 for off-target in both cases; PSSM actively rejects off-target sequences).

Channel	Substrate	AUROC	95 % CI	Confidence
MMP-9	`IPRTLTAA`	0.837	[0.820, 0.853]	**HIGH**
Neutrophil ser. pr.	`AAPVSGIG`	0.899	[0.877, 0.919]	**HIGH**